UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026). Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	FALSE	A plotter is an input device.	A plotter produces output.
A2	TRUE	Sampling converts a continuous sound wave into digital values.	Sampling measures the wave at intervals.
A3	FALSE	Anti-aliasing makes the edges of text look more jagged.	Anti-aliasing smooths jagged edges.
A4	TRUE	The amplitude of a sound wave determines its loudness.	Larger amplitude means louder sound.
A5	TRUE	Streaming allows media to begin playing before the whole file is downloaded.	Streaming delivers media continuously.
A6	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.
A7	TRUE	Multimedia combines text, graphics, audio, video and animation.	These are the core multimedia elements.
A8	TRUE	Kerning adjusts the space between individual character pairs.	Kerning changes spacing for a specific pair.
A9	TRUE	Authoring tools help developers organise and edit multimedia elements.	Authoring tools assemble media.
A10	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A11	TRUE	CMYK is used for printed output.	CMYK is a subtractive print model.
A12	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A13	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.
A14	TRUE	Hertz measures sound frequency.	Hertz is cycles per second.
A15	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.

© Hak Cipta Universiti Teknologi MARA

PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	M	Copyright	exclusive rights of a creator
B2	E	Scanner	input device that digitises documents
B3	H	Keyframe	important frame in an animation
B4	N	Serif	font with small strokes at character ends
B5	I	Bitmap	image made of pixels
B6	C	Tweening	generating in-between frames
B7	J	Cel animation	hand-drawn frame-by-frame animation
B8	A	Multimedia	integration of text, graphics, audio, video and animation
B9	B	Linear multimedia	presentation without user control of sequence
B10	O	Storyboard	visual sequence plan
B11	F	Frame rate	frames displayed per second
B12	L	Vector	image made of mathematical objects
B13	D	Virtual reality	simulated environment
B14	K	Card-based tool	authoring tool organised like pages or cards
B15	G	Chroma key	replacing a colour background

© Hak Cipta Universiti Teknologi MARA

PART C - ANSWER KEY

Q	Ans	Correct answer
C1	C	Multimedia with hyperlinks
C2	C	Printed output
C3	C	Messages with a beginning and end
C4	A	Vector
C5	B	Text, graphics, audio, video and animation
C6	C	Plotter
C7	D	Loudness
C8	A	Non-linear
C9	C	Improves accuracy
C10	C	Along a predefined route
C11	B	Display web pages
C12	B	Card-based
C13	D	One shape into another
C14	C	It combines visuals and sound
C15	A	CMYK
C16	C	A cinema presentation
C17	D	Another resource
C18	A	Screen display
C19	B	Assemble multimedia elements
C20	B	The user can control content and flow
C21	A	Analog sound to digital values
C22	B	Drawing input
C23	C	A low-fidelity layout sketch
C24	A	Play while downloading
C25	D	Replace a colour background
C26	C	Structure web content
C27	C	Important frames in an animation
C28	C	Draws alternate lines of a frame
C29	C	PNG
C30	A	Automatic device recognition

© Hak Cipta Universiti Teknologi MARA

PART D - WORKED ANSWERS

QUESTION 1

Topic: web hypermedia

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

Linear follows a fixed sequence; non-linear lets users choose their own path.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

They connect related exhibits and let users move between pages.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

Compression reduces file size for faster transfer; streaming lets audio or video play while it downloads.

(4 marks)

QUESTION 2

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(3 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

QUESTION 3

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(5 marks)

© Hak Cipta Universiti Teknologi MARA

QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA